

SentinelX

AI Architecture Explanation

Inputs (What Data Goes In)

SentinelX has three distinct intake mechanisms, each feeding a different part of the system, and only two of which involve AI at all.

- SOS: a 3-second long-press on the home screen's SOS button. No data is analyzed by AI — this immediately triggers an alert to the user's emergency contacts and the NGO dashboard.
- Detect Threat: when toggled on, the mobile app captures short segments of audio, transcribes them to text on the device, and immediately discards the audio. Only the transcribed text is used as input to the on-device AI — raw audio never leaves the phone.
- Safe Chat + Upload Evidence: a guided 4-step structured intake (category, narrative, timing, frequency), followed by an optional upload of video, audio, or image evidence documenting the incident.

The Safe Chat intake workflow specifically collects:

- contextual incident categories,
- free-text narrative descriptions,
- timeline indicators,
- escalation frequency patterns,
- and optionally, uploaded video, audio, or image evidence.

Step 1 — Contextual Category Selection

Users select the type of scenario they are experiencing, including: stalking, cyber-harassment, threatening communication, physical tracking, coercion, or contextual exploitation patterns.

Step 2 — Narrative Description

Users describe the situation in free-text form using their own words.

Step 3 — Timeline Context

Users specify when the incidents occurred: today, this week, or months ago.

Step 4 — Escalation Frequency

Users indicate whether the situation happened once, occurred repeatedly, or demonstrates ongoing escalation behavior.

Step 5 — Upload Evidence

Users may optionally attach video, audio, or image evidence documenting coercion, sextortion, grooming, or other abuse described in their report. This evidence — not the narrative text — is what the Gemini API later analyzes.

Importantly, all incoming Safe Chat reports are initially stored securely as static, un-triaged records before any AI processing occurs. No automatic AI analysis is performed upon submission. This architecture was intentionally designed to preserve human oversight and reduce the risks associated with autonomous threat interpretation in sensitive safety environments.

AI Capability Used

SentinelX uses two independent AI components, deployed in two completely different places, each scoped to one job:

- A Bidirectional Long Short-Term Memory (Bi-LSTM) model, deployed directly on the user's device, performing threat / non-threat language classification on locally-transcribed text from the Detect Threat feature. It never sees or stores raw audio, and it never takes action — it only flags.
- The Gemini API, running on the NGO dashboard side, which analyzes uploaded video, audio, or image evidence from Safe Chat reports — but only after an analyst explicitly requests it.

SOS uses no AI capability at all — it is a direct alert mechanism by design, kept deliberately simple for time-critical emergencies.

AI was necessary for the other two flows because contextual exploitation, coercion, stalking, and abuse patterns cannot be reliably identified using static keyword systems or traditional rule-based logic alone. Spoken threats depend on repetition, timing, and framing that a keyword filter would miss — which is why Detect Threat uses a trained language model instead. Evidence such as

video or audio of coercion or grooming requires contextual interpretation of unstructured media — which is why Gemini, not a rules engine, analyzes it.

What Processing Happens

Because SentinelX has three distinct flows, each has its own processing path and its own type of human control point.

Flow 1 — SOS (No AI)

A 3-second long-press on the SOS button immediately sends an alert to the user's emergency contacts and the NGO dashboard. There is no AI processing step, no delay, and no analysis — by design, for genuine emergencies.

Flow 2 — Detect Threat (On-Device, Consent-Gated)

The human control point here is the user's own toggle — Detect Threat only runs while intentionally switched on. While active: audio is captured, transcribed to text locally, and the audio is immediately discarded. The transcribed text is classified by the on-device Bi-LSTM as threat or non-threat. If classified as a threat, only the classification result and the relevant transcript snippet are sent to the NGO dashboard for analyst review. The model never escalates, contacts authorities, or acts on the user's behalf — it only flags.

Flow 3 — Safe Chat + Upload Evidence (Analyst-Gated)

Step 1 — Intake Storage: When a user submits a Safe Chat report with optional evidence, it is stored inside MongoDB as a static case record with status “Pending Analysis.” No AI inference occurs at this stage.

Step 2 — Human Trigger: An authorized NGO analyst reviews the queue and intentionally clicks [Analyze Report] for a specific case. This click is the mandatory human authorization checkpoint — human agency remains the sole catalyst for evidence analysis.

Step 3 — Gemini Evidence Analysis: After authorization, the Flask backend sends the uploaded evidence — video, audio, or image — to the Gemini API, which analyzes the evidence and returns a plain-language, explainable contextual analysis of what it shows.

Step 4 — Explainable Output: The report status transitions from “Pending Analysis” to “Triaged,” and the dashboard surfaces Gemini’s evidence analysis alongside structured case context for the analyst.

Multimodal Evidence Synthesis Workflow

For Safe Chat investigations, SentinelX also supports multimodal contextual evidence analysis through Gemini-powered synthesis workflows.

When a vulnerable user submits a Safe Chat report, they may optionally attach supporting evidence such as:

- Screenshots
- Chat exports
- Voice notes
- Images
- Short videos related to coercion, grooming, cyber-harassment, sextortion, stalking, or exploitation incidents

Importantly, these uploads are not automatically analyzed upon submission.

The evidence remains in a **Pending Analysis** state until an authorized NGO analyst intentionally initiates a review from the dashboard interface.

Once the analyst clicks **Analyze Report**, the Flask backend retrieves:

- Narrative Description
- Timing Metadata
- Frequency Indicators
- Uploaded Evidence

The system then securely routes this contextual payload into the Gemini multimodal synthesis pipeline.

The AI workflow evaluates:

- Contextual behavioral framing
- Escalation patterns
- Intimidation progression

- Grooming indicators
- Coercive interaction structures
- Cross-modal consistency between uploaded evidence and written narratives

The synthesis engine can also assist analysts in identifying probable communication environments or digital platforms associated with the abuse context by analyzing recurring interface structures, conversational framing, and contextual interaction signals contained within uploaded evidence.

Rather than producing autonomous conclusions, the system generates explainable contextual synthesis outputs designed to help investigators transform fragmented reports into structured investigative intelligence.

The resulting synthesis is returned to the NGO dashboard as an Explainable AI investigative report containing:

- Contextual risk observations
- Behavioral escalation indicators
- Narrative synthesis summaries
- Transparent reasoning outputs for human review

Outputs (What the User Receives)

Authorized organizational analysts receive three distinct types of output, depending on which flow generated them:

- an instant SOS alert with emergency contact details (no AI involved),
- a Detect Threat flag from the on-device Bi-LSTM, including the relevant transcript snippet and threat / non-threat classification,
- a Gemini-generated contextual analysis of uploaded evidence (video, audio, or image) for an analyst-triggered Safe Chat report,
- plus structured case summaries, escalation indicators, and triaged organizational intelligence views combining all three.

Importantly, SentinelX does not autonomously contact authorities, escalate cases, trigger interventions, or make final safety decisions in any of the three flows. Human analysts remain fully responsible for interpreting all signals, validating insights, determining intervention necessity, and making all real-world escalation decisions.